



Koneru Lakshmaiah Education Foundation

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STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	I-III Semester (2025-26)
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Roll Number	2520090202
Branch	CSIT
Course Outcome	CO2

Case Study Topic: Foxconn Factory Floor – Segment Tree for Sensor Range-Max Alerts

Work Done Summary:

In this case study, a Segment Tree was implemented to process temperature sensor readings efficiently. The system supports range maximum queries and point updates for real-time monitoring applications.

Implementation Details:

- Created an array to store sensor temperature readings.
- Constructed a Segment Tree using recursive tree building.
- Stored maximum values in internal tree nodes.
- Implemented range maximum query functionality.
- Handled complete, partial, and no-overlap cases.
- Performed point update operations on sensor readings.
- Updated affected tree nodes after modification.
- Displayed query results before and after updates.

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Result: Screenshots/Output

Output

```
FACTORY SENSOR ALERT SYSTEM
```

```
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```

```
Maximum Temperature [3..7] = 82
```

```
After Update (75 -> 88)
```

```
Maximum Temperature [2..6] = 88
```

```
=== Code Execution Successful ===
```

Observations:

Segment Tree was successfully constructed from sensor data.

- Range maximum queries returned correct results.
- Point updates propagated correctly to parent nodes.

- Query operations required fewer comparisons than linear search.
- Update operations were completed efficiently.
- The system supported real-time sensor monitoring.

Hub Link: <https://github.com/gowthampenmatsa07-prog/D-S-A.git>

Student Signature	Faculty Signature